

Homework

Preliminary Mathematics Standard

Managing Money

1 Questions

1.1 Purchasing Goods

1. A salesperson's salary is \$67 800, and she is offered an increase of 5% for 2 years, or 2% for the first year and 3% for the second year. Which option will give her the most increase in her salary after the 2 years? Justify your answer using mathematical calculations.
2. A boat is bought for \$4 500 and sold 3 years later for a 45% profit. Determine the selling price of the boat. Hence, explain the method to determine the selling price of an item when given the percentage profit or loss.
3. To determine the price of an item before GST was added, the shop owner calculates 10% of the item and then subtracts this amount from the selling price. Explain, using mathematical calculations, why this method will not find the price of items before GST is added.
4. A loan is an investment in reverse; you borrow money from a bank and are charged interest. The value of a loan becomes its total cost.
 - (a) A worker wishes to borrow \$10 000 from a bank, which charges 11.5% interest per year. If the loan is over 2 years:
 - (i) calculate the total interest paid
 - (ii) calculate the total cost of the loan.
 - (b) Most loans require a monthly payment. The monthly payment of a simple interest loan is calculated by dividing the total cost by the number of payments made during the term of the loan. Determine the monthly payment for the loan in part (a) above.
5. A family has both electricity and gas connected to their house. Their average electricity usage for 90 billing days is 987 kWh and their average gas usage is 1 650 MJ per month (30 days). Service to property charges are the same for both providers. The family decides to change providers. They have the following choices:

Choice 1: Separate energy providers

- Electricity: all day rate 10.03 c/kWh
- Pay on time discount (electricity only): 12.5%
- Gas: all day rate 1.89 c/MJ

Choice 2: Bundle both electricity and gas with one provider; receive the following discounts

- 17% discount on gas
- 10% discount on electricity
- Electricity daily rate: 12.01 c/kWh
- Gas daily rate: 2.03 c/MJ

Which choice will be the most cost-effective for the family, assuming they always pay on time?

6. A council plans to raise \$7.5 million in revenue. The capital improved value of home owners in the council area is \$2.5 billion. To calculate the rate in the dollar, the council performs the following calculation:

$$\frac{7.5}{2.5} = 3$$

The capital improved value of a particular home is \$175 000. To calculate the council rates owing for that home, a council worker performs the following calculation:

$$\frac{3}{100} \times 175\,000 = \$5\,250$$

The homeowner complains that the rates are far too high. Is there an error in the calculation performed by the council and/or by the council worker? Justify your answer by calculating the council rates for the rate payer's property.

1.2 Budgeting

1. A vehicle is purchased for \$24 980 through finance that attracts 6% interest for 5 years. The owner pays a 10% deposit and then pays \$496.28 monthly. This value for the monthly repayments includes a \$250 one-off administration fee to set up the loan and a monthly \$5 loan-servicing fee. Show that the comparison rate for the loan is 6.489% per annum.
2. A customer has a choice between two electricity plans, A and B, shown below.

	Plan A	Plan B
	Charge/rate	Charge/rate
Off-peak (11 pm – 7 am)	12.03 c/kWh	9.08 c/kWh
Peak	12.03 c/kWh	13.01 c/kWh
Service to property	\$1.01/day	\$0.99/day
Pay on time discount	15%	12%

The customer's energy usage over 90 billing days is 534 kWh (peak time) and 757 kWh (off-peak time). Which plan would be the most cost-effective for the customer? Justify your answer by calculating the electricity costs for each plan if the customer pays on time and if he does not pay on time.

3. A female 19-year-old driver has three options to purchase comprehensive car insurance for a small car that will be parked in a locked garage in a suburban property. All options include replacing her car, providing transportation if the car is unable to be driven, and costs for injuries for all passengers.

- **Option A:** \$1 143, \$850 excess, 10% online discount
- **Option B:** \$1 149, \$850 excess
- **Option C:** \$1 155, \$800 excess, 15% online discount

The excess fee is the amount paid to the insurer if a claim is made. Which option is the most cost-effective?

4. The fuel consumption for a vehicle travelling in city traffic is recorded as 9.8 L/100 km. When travelling along highways, the fuel consumption is recorded as 7.5 L/100 km.

- (a) Show that the average fuel consumption is 8.65 L/100 km.
- (b) The vehicle travels 126 km along city roads and 458 km along the highway. Calculate the total number of litres of fuel used, correct to 1 decimal place.
- (c) The cost of fuel is \$1.25 per litre. Calculate the actual cost of fuel, correct to the nearest cent.
- (d) When calculating the fuel costs using the average fuel consumption, the value is different to the actual cost. Explain why.

5. A family's net annual income is \$69 480. Their spending is shown in the following table.

Item	Frequency	Amount
Holiday/entertainment	Yearly	\$4 500
Mortgage	Weekly	\$245
Insurance	Yearly	\$1 980
Car costs	Monthly	\$950
Food	Weekly	\$145
Electricity, water	Quarterly	\$580
School fees	Yearly	\$8 500
Health insurance	Monthly	\$368

- (a) Prepare a monthly budget and show whether the family is living within their means (i.e. making a profit).
 - (b) The family needs to adjust their spending. By identifying the fixed and discretionary spending items, explain which items they should consider reducing.
6. A person's monthly spending is \$1 850. His current spending is 55% of his net monthly income.
 - (a) Determine his net monthly income and net annual salary.
 - (b) If he gets paid an hourly rate of \$22.50, how many hours a week does he work? Give your answer correct to 1 decimal place.
 - (c) He gets a job that pays 4% more than he is currently earning. If his spending does not change, what percentage will his spending be of his net monthly income? Give your answer correct to 2 decimal places.
7. A loan is an investment in reverse; you borrow money from a bank and are charged interest. The value of a loan becomes its total cost.
 - (a) A worker wishes to borrow \$10 000 from a bank, which charges 11.5% interest per year. If the loan is over 2 years:
 - (i) calculate the total interest paid
 - (ii) calculate the total cost of the loan.
 - (b) Most loans require a monthly payment. The monthly payment of a simple interest loan is calculated by dividing the total cost by the number of payments made during the term of the loan. Determine the monthly payment for the loan in part (a) above.

8. A family household's bills for 3 months (1 quarter) are shown.

Electricity bill

Tariff	Bill days	Current reading	Previous reading	Total usage	Rate (c/kWh)
Peak	90	12 589	11 902		31.08
Off-peak	90	2 058	1 511		14.05
Service to property	90				\$1.075/day

Water bill

Bill details as of 31 March	Value	Price
Water service charge	1 Jan to 31 Mar	22.51
Sewerage service charge	1 Jan to 31 Mar	145.90
Storm water	1 Jan to 31 Mar	18.70

	Volume (kL)	Charge/kL	Price
Water usage	55	\$2.00/kL	
Recycled water usage	35	\$1.79/kL	

Council rates

Capital improved value \$280 000, rate in the dollar 0.008.

- Show that the quarterly council rate installment is \$560.
 - Calculate the amount that the family spends on electricity, water and council rates for the quarter. (Remember to include GST where relevant.)
 - The family's monthly net income is \$6 019.35 (after tax). Calculate the percentage of their income that goes towards paying the household bills. Give your answer correct to 2 decimal places.
 - If the family wishes to reduce their household bills, what advice would you give?
9. A family of four are going on an overseas holiday for 6 weeks. They have saved \$20 000 for the trip and recorded the costs for the essentials but do not know how much to spend on food.

Item	Cost
Passports	\$832
Flights	\$5 668
Travel insurance	\$658
Accommodation	\$245 per night
Food	

- (a) They estimate food costs to be about \$45 per day. How much money do they have remaining from the \$20 000 to spend on their trip?
- (b) The family reassesses and decides to adjust their spending to save more money for their trip. Their monthly spending is \$2 585 and net monthly income is \$4 580. If they can save half of their monthly profit for 4 months, how much more money will they have for their overseas trip?

2 Answers

2.1 Purchasing Goods

1. Option 2 salary after two years is \$71230.68.
Option 1 salary after two years is \$71190.
Therefore option 2 is the better option.
2. Selling price = \$6 525.
3. Price including GST is 10% more than the original price (i.e. 110%).
Finding 10% of the selling price gives 10% of 110 = 11. Subtracting 11% from the selling price gives $110 - 11 = 99\%$, which is less than the original price.
4. (a) (i) Total interest = \$2 300
(ii) Total cost = \$12 300
(b) Monthly payment = \$512.50
5. Choice 1
6. The council's rate in the dollar calculation is incorrect — the units should be cents in the dollar (i.e. $\frac{7\,500\,000}{2\,500\,000\,000} = 0.003$ dollars, or 0.3 cents in the dollar). The correct rates for the \$175 000 property are $0.003 \times 175\,000 = \525 , not \$5 250.

2.2 Budgeting

1. Comparison rate $\approx 6.489\%$ p.a.
2. Plan B is the better option for both when he pays on time (\$220.03 compared to \$230.20) and when he does not (\$250.05 compared to \$270.83).
3. Option A (after 10% discount): \$1 028.70
Option B (no discount): \$1 149.00
Option C (after 15% discount): \$981.75
Option C is the most cost-effective (lowest premium). However, it has the lowest excess (\$800 vs \$850), which is a minor advantage if a claim is made.
4. (a) Average = $(9.8 + 7.5) \div 2 = 8.65$ L/100 km ✓
(b) City: $9.8 \times 1.26 = 12.348$ L; highway: $7.5 \times 4.58 = 34.35$ L; total = 46.7 L
(c) $46.698 \times 1.25 = \$58.37$
(d) The average fuel consumption is a value that assumes equal distance (1 : 1) of driving under the two conditions (city and highway). In this case, the ratio of city to highway driving was 126 : 458 (1 : 3.6). This

means the overall fuel consumption is lower than the average value would predict, because 3.6 times more driving took place at the lower fuel consumption.

5. The items they should consider reducing are the discretionary items, particularly the non-essential items: holidays and entertainment. These items do not have fixed costs, which means that the costs can be adjusted based on usage or need.

(a) See table below.

Item	Frequency	Amount	Monthly cost
Holiday/entertainment	Yearly	\$4500	\$375.00
Mortgage	Weekly	\$245	\$1061.67
Insurance	Yearly	\$1980	\$165.00
Car costs	Monthly	\$950	\$950.00
Food	Weekly	\$145	\$628.33
Electricity, water	Quarterly	\$580	\$193.33
School fees	Yearly	\$8500	\$708.33
Health Insurance	Monthly	\$368	\$368.00
Total			\$4449.67
Income	Yearly	\$69 480	\$5790.00
Profit			\$1340.33

(b) See table below.

Fixed	Discretionary
Mortgage	Holiday/entertainment
Insurance	Car costs
Health insurance	Food
School fees	Electricity, water

6. (a) Net monthly income = $\$1\,850 \div 0.55 = \$3\,363.64$; net annual salary = $\$40\,363.64$
 (b) Hours per week = $\$40\,363.64 \div (22.50 \times 52) \approx 34.5$ hours
 (c) New monthly income = $\$3\,363.64 \times 1.04 = \$3\,498.18$; spending percentage = $\$1\,850 \div \$3\,498.18 \times 100 \approx 52.88\%$
7. (a) (i) Interest = $\$10\,000 \times 0.115 \times 2 = \$2\,300$
 (ii) Total cost = $\$12\,300$
 (b) Monthly payment = $\$12\,300 \div 24 = \512.50
8. (a) $0.008 \times \$280\,000 = \$2\,240$ annually; quarterly = $\$2\,240 \div 4 = \560 ✓

(b) \$1345.59

(c) 22.35%

(d) Answers may vary but could include the following.

The family could reduce their electricity bill by installing energy-saving light bulbs and turning off lights and power appliances at the power point. They could reduce their water usage by ensuring there are no leaking taps, reducing time spent in the shower, washing full loads of washing, and ensuring the dishwasher is full before turning it on.

Council rates are set and will not be reduced by changing household behaviours.

9. (a) \$662

(b) Monthly profit = $\$4\,580 - \$2\,585 = \$1\,995$; half = $\$997.50$; over 4 months = $\$3\,990$